Chapter 18 Notes

Section 2

Lemma 2.2

Here is why $\cup_{q_0 \in \mathcal{G}} S(q_0)$ is an M-set (and hence so is S). Let $G(x, q)$ stand for the formula

$$(\forall y \in x)(\text{Ordinal}(y) \wedge (\exists g \in R_\beta^{\mathcal{G}})(q \Vdash [\![\hat{y} \approx g]\!]))$$

(this can be expanded to a real first-order formula by results in the previous chapter). If $S(q_0)$ happens to be in M, then $G(S(q_0), q_0)$ will be true in M. Now, we already know that each $S(q_0) \in M$. Then consider the following function which is definable over M:

$$x \in \mathcal{G} \wedge G(y, x)$$

Because the domain, $\mathcal{G}$, is in M, so is the range, and because we have shown each $S(q_0)$ to be in M, the range is the set of all $S(q_0)$ (for $q_0 \in \mathcal{G}$). Then the union of this, which is also in M, is just S itself.

Theorem 2.3

It is not fully clear to me why $p \Vdash [\![(\text{Ordinal}(f) \wedge \text{Ordinal}(\hat{\delta})) \supset (f \in \hat{\delta} \vee \hat{\delta} \in f \vee f \approx \hat{\delta})]\!]$.

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.

Section 3

Theorem 3.2

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.

Section 4

Notes

It should be noted why both $\mathcal{G} \in M$ and $\mathcal{R} \in M$ (which are both conditions we need to have). $\mathcal{G}$ is a subset of $(\mathcal{P} \cap M) \times (\mathcal{P} \cap M)$, which is in M. Also, $\mathcal{G}$ is definable over M by

$$(\exists P)(\exists N)(\exists n)(\exists m)(\exists f)(\exists g)(x = \langle P, N \rangle \wedge \text{Num}(n) \wedge \text{Num}(m) \wedge$$

$$\text{f is 1-1 func}(n, N) \wedge \text{g is 1-1 func}(m, P) \wedge P \cap N = \emptyset)$$

(this works because, for each 1-1 function from a number n to some N or P, it is finite, so it is in M). Thus $\mathcal{G} \in M$. As for $\mathcal{R}$, it is a subset of $\mathcal{G} \times \mathcal{G}$, which is in M, and is defined over M by the formula:

$$(\exists P_1, P_2)(\exists N_1, N_2)(x = \langle P_1, N_1\rangle \wedge y = \langle P_2, N_2\rangle \wedge x \in \mathcal{G} \wedge y \in \mathcal{G}$$

$$\wedge P_1 \subseteq P_2 \wedge N_1 \subseteq N_2)$$

and thus $\mathcal{R} \in M$.

In Case 1 of Theorem 4.1, note that $r' \vDash [\![\sim(\hat{n} \in \hat{c})]\!]$ would follow by Proposition 3.3 of the previous chapter.

Exercise 4.1

Suppose $n \in N$. We show that $\langle P, N\rangle \vDash [\![\sim(\hat{n} \epsilon f_0)]\!]$. Because $n \in N$, $n \notin P$. Thus $\langle P, N\rangle \vDash \sim(\hat{n} \epsilon f_0)$. Now, if $\langle P, N\rangle \mathcal{R} \langle P_1, N_1\rangle$, then $N \subseteq N_1$, and thus $n \in N_1$, but $n \notin P_1$, which would mean $\langle P_1, N_1\rangle \vDash \sim(\hat{n} \epsilon f_0)$. This explains why $\langle P, N\rangle \vDash \Box\sim(\hat{n} \epsilon f_0)$, and thus $\langle P, N\rangle \vDash [\![\sim(\hat{n} \epsilon f_0)]\!]$.

Section 5

Notes

Here is why the negation of the formula on page 253 is Σ formula. Its negation is

$$\sim(\text{Ordinal}(\alpha) \wedge (\forall \beta \in \alpha)\sim(\exists f)(\text{1-1 onto}(f, \beta, \alpha)))$$

$$\sim\text{Ordinal}(\alpha) \vee \sim(\forall \beta \in \alpha)\sim(\exists f)(\text{1-1 onto}(f, \beta, \alpha))$$

$$\sim\text{Ordina}(\alpha) \vee (\exists \beta \in \alpha)(\exists f)(\text{1-1 onto}(f, \beta, \alpha))$$

and this is Σ because Ordinal(α) and 1-1 onto(f, β, α) are both Σ_0.

In Lemma 5.3, it is not clear to me what "is" stands for. Previously, in section 3 of this chapter, it was said that y is L_x stands for the formula $\mathcal{M}(x, y)$ defined in section 4 of Chapter 14. Clearly that is not what "is" stands for in this section. Whatever the case, suppose that [x is $f(y) \wedge x'$ is $f(y)$] $\supset x = x'$ is indeed classically valid; here is why this means $r \vDash [\![\hat{d}_1 \approx \hat{d}_2]\!]$. Using standard $S4$ manipulation, it follows that $r \vDash [\![\hat{d}_1 = \hat{d}_2]\!]$. By Theorem 2.10 of Chapter 17, this means that $p \vDash [\![\hat{d}_1 \approx \hat{d}_2]\!]$.

Also, in Lemma 5.3, note why the choice of q works. That is, when, for each $d \in F(c)$, we associate some q with it such that $p\mathcal{R}q$. What we need of this association is that, for each $d \in F(c)$, the q associated with d is unique (if we didn't have this, we couldn't conclude that $F(c)$ could be placed in a 1-1 correspondence with the set of all associates, which we need to show that $F(c)$ must be countable). Well, if, for some $d_1, d_2 \in F(c)$, we had the q_1 and q_2 associated, respectively, with each of d_1 and d_2, such that $q_1 = q_2$, it would trivially follow that q_1 and q_2 were compatible. Then, by the result in the second paragraph of the proof, it would have to be that $d_1 = d_2$.

Lemma 5.4

Here is why, in the second paragraph, we can conlclude that $p \vDash \Diamond[\![\hat{\delta} \text{ is } f(\hat{\gamma})]\!]$ for some $\gamma \in \beta$. We know that $p \vDash [\![(\exists y)(y \in \hat{\beta} \wedge \hat{\delta} \text{ is } f(y))]\!]$, meaning, at some world p' accessible from p, there is some d such that $p' \vDash [\![d \in \hat{\beta}]\!]$ and $p' \vDash [\![\hat{\delta} \text{ is } f(d)]\!]$. Because $p' \vDash [\![d \in \hat{\beta}]\!]$, $p'' \vDash [\![d \approx e]\!]$ and $p'' \vDash e \epsilon \hat{\beta}$ for some p'' with $p'\mathcal{R}p''$. But this means e is $\hat{\gamma}$ for some $\gamma \in \beta$. Also, $p'' \vDash [\![\hat{\delta} \text{ is } f(d)]\!]$, and by substitutivity, it follows that $p'' \vDash [\![\hat{\delta} \text{ is } f(\hat{\gamma})]\!]$. Thus, because $p\mathcal{R}p''$, $p \vDash \Diamond[\![\hat{\delta} \text{ is } f(\hat{\gamma})]\!]$ for some $\gamma \in \beta$.

Note that, in general, it does not follow that, if $f \in \mathcal{D}^{\mathcal{G}}$, then $f = \hat{x}$ for some $x \in M$. This is because, if this were the case, f would need to have ordered pairs of the form $\langle p, y\rangle$ (for some y) for every possible world p, and this certainly is not the case for all $f \in \mathcal{D}^{\mathcal{G}}$.

It should be noted why Lemma 5.4 does not immediately follow from Proposition 3.3 of the last chapter. For instance, one might argue that, by that result of the last chapter, $p \vDash [\![\text{1-1 onto}(f, \hat{\beta}, \hat{\alpha})]\!]$ would immediately imply 1-1 onto(f, β, α) being true in M, a contradiction to the asusmption that $\beta < \alpha$ and α is a cardinal. However, this cannot be concluded - to do that we would need to know that $f = \hat{x}$ for some $x \in M$, which we do not know.

Theorem 5.5

Note why $p' \vDash [\![b \in \hat{\alpha}]\!]$ implies $p'' \vdash [\![b \approx \hat{\beta}]\!]$ for some p'' with $p'\mathcal{R}p''$ and $\beta \in \alpha$. Because $p' \vDash [\![b \in \hat{\alpha}]\!]$, $p'' \vDash [\![b \approx x]\!]$ and $p'' \vDash [\![x \epsilon \hat{\alpha}]\!]$ for some p'' with $p'\mathcal{R}p''$. Then x is some $\hat{\beta}$ for $\beta \in \alpha$, and thus $p'' \vDash [\![b \approx \hat{\beta}]\!]$.

There is an error in this proof. When the contradiction arises with $p'' \vDash [\![\text{1-1 onto}(f, \hat{\beta}, \hat{\alpha})]\!]$, by using Lemma 5.4, this inference is not actually correct. For, it could be that β is finite - yet Lemma 5.4 requires β to be infinite. Whether or not Lemma 5.4 can be proven if β is finite is not fully clear to me - it depends upon what the relation "is" means, but this has not been defined. If we suppose that $p \vDash [\![x \text{ is } y]\!]$ implies $p \vDash [\![x \approx y]\!]$, then Lemma 5.4 certainly follows if β is finite.

Proof. We show the result of Lemma 5.4, but in the case that β is finite. This is actually really easy - we don't even need to define the function F. We aim to derive a contradiction by showing that β must contain an infinite number of elements. Suppose that β contained n elements. Because α is infinite, we can choose $n+1$ (distinct) elements from it, say $a_1, ..., a_n, a_{n+1}$. Because $p \vDash [\![\text{1-1 onto}(f, \hat{\beta}, \hat{\alpha})]\!]$, we know that $p \vDash [\![(\hat{a_i} \in \hat{\alpha} \supset (\exists y)(y \in \hat{\beta} \wedge \hat{a_i} \text{ is } f(y)))]\!]$ for each a_i (for $i \leq n+1$). It can easily (by induction on n) be seen that this means there exist $c_1, c_2, ..., c_n, c_{n+1} \in \beta$ and some p' with $p\mathcal{R}p'$ such that $p' \vDash [\![\hat{c_i} \in \hat{\beta}$ and $p' \vDash [\![\hat{a_i} \text{ is } f(\hat{c_i})]\!]$ (for each $i \leq n+1$). Because β was assumed to have only n members, it must be that $c_i = c_j$ for some i and j. Then $p' \vDash [\![\hat{c_i} \approx \hat{c_j}$, meaning $p' \vDash [\![f(\hat{c_i}) \approx f(\hat{c_j})]\!]$. Because we have both $p' \vDash [\![\hat{a_i} \text{ is } f(\hat{c_i})]\!]$ and $p' \vDash [\![\hat{a_j} \text{ is } f(\hat{c_j})]\!]$, and we are assuming $p' \vDash [\![x \text{ is }]\!]$ implies $p' \vDash [\![x \approx y]\!]$ (we are assuming that this is what "is" measn), it follows by transitivity that $p' \vDash [\![\hat{a_i} \approx \hat{a_j}]\!]$, meaning $p' \vDash [\![\hat{a_i} = \hat{a_j}]\!]$. This means $a_i = a_j$, contrary to the assumption that a_i and a_j are distinct.

Thus, no matter what size β is (assuming β is finite), assuming $p \vDash [\![\text{1-1 onto}(f, \hat{\beta}, \hat{\alpha})$ led to a contradiction, therefore completing the proof. □

Exercise 5.1

We show that $\aleph_0 \times \overline{\overline{A}} \geq \overline{\overline{\cup A}}$. By Lemma 5.3, $\aleph_0$ is an upper bound on the size for members of A. Then an upper bound on the size of $\overline{\overline{\cup A}}$ is obviously $\aleph_0 \times \overline{\overline{A}}$ (because in the case of the upper bound, it could be 1-1 mapped onto $\aleph_0 \times \overline{\overline{A}}$) from whence it follows that $\aleph_0 \times \overline{\overline{A}} \geq \overline{\overline{\cup A}}$.

Section 6

Lemma 6.1

Note that, if $p \vDash [\![x \approx x']\!]$, it follows that $p \vDash [\![x = x']\!]$ (we prove this below). This explains why $\gamma_1 \neq \gamma_2$ implies $[\![\sim(\hat{\gamma_1} \approx \hat{\gamma_2})]\!]$ being valid in the $S4$ model.

Proof. We prove that, if $p \vDash [\![f \approx f']\!]$ (for some $f, f' \in \mathcal{D}^{\mathcal{G}}$), then $p \vDash [\![f = f']\!]$. But suppose not. That is, $p \nvDash [\![f = f']\!]$. Then either (1) $p \nvDash [\![(\forall x)(x \in f \supset x \in f')]\!]$ or $p \nvDash [\![(\forall x)(x \in f' \supset x \in f)]\!]$. Without loss of generality, suppose that it is the former case. Now, because $p \vDash [\![f \approx f']\!]$, we know that (2) $p \vDash [\![(\forall x)[(x \epsilon f) \supset (\exists y)(y \epsilon f' \wedge y \approx x)]]\!]$ (by Proposition 2.12 of Chapter 17). By (1), there is some d and some p' with $p\mathcal{R}p'$ such that $p' \vDash [\![d \in f]\!]$ and $p' \vDash [\![\sim(d \in f')]\!]$. We derive a contradiction by showing that $p' \vDash [\![d \in f']\!]$. We do this by showing $p' \vDash \Box\Diamond(d \in f')$.

Consider any p'' with $p'\mathcal{R}p''$. Because $p' \vDash [\![d \in f]\!]$, there is some p''' with $p''\mathcal{R}p'''$ and some b such that $p''' \vDash [\![b \epsilon f]\!]$ and $p''' \vDash [\![d \approx b]\!]$. By (2), it follows that there is some c and some q with $p'''\mathcal{R}q$ such that $q \vDash [\![c \epsilon f']\!]$ and $q \vDash [\![c \approx b]\!]$. Because $p'''\mathcal{R}q$, $q \vDash [\![d \approx b]\!]$, meaning, by transitivity, $q \vDash [\![d \approx c]\!]$, and thus, by definition of $\in$, $q \vDash d \in f'$. Hence, because $p''\mathcal{R}q$, $p'' \vDash \Diamond(d \in f')$. Because p'' was arbitrary, it follows that $p' \vDash \Box\Diamond(d \in f')$, that is, $p' \vDash [\![(d \in f')]\!]$, a contradiction. □